COST MANAGEMENT PLAN

LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

ASIA PACIFIC COLLEGE
3 HUMABON, MAGALLANES
MAKATI CITY, METRO MANILA 1232

APRIL 1, 2026

TABLE OF CONTENTS

INTRODUCTION 3
COST MANAGEMENT APPROACH 3
PROJECT ASSUMPTIONS & SPRINT CONFIGURATION 4
a. Sprint Assumptions 4
SPRINT BUDGET DETAILS 4
a. Planned Labor Costs 4
b. Planned Unit Costs 5
c. Actual Unit Costs — Co-working Space (Hourly) 6
MEASURING PROJECT COSTS 7
a. Overview 7
b. EVM Computation — Term 3 Sprint 1 7
c. EVM Interpretation 8
COST PERFORMANCE THRESHOLDS 8
REPORTING FORMAT 9
COST VARIANCE RESPONSE PROCESS 9
COST CHANGE CONTROL PROCESS 10
PROJECT BUDGET SUMMARY 10

INTRODUCTION

The Cost Management Plan clearly defines how the costs of this project will be managed throughout the project's lifecycle. It sets the format and standards by which project costs are measured, reported, and controlled.

Specifically, this plan:

- Identifies who is responsible for managing costs
- Identifies who has the authority to approve changes to the project or its budget
- Defines how cost performance is quantitatively measured and reported
- Establishes report formats, frequency, and the audience for each report

The Project Manager is responsible for managing and reporting on the project's cost throughout the duration of the project. During sprint reviews and monthly project status meetings, the Project Manager will present and review the project's cost performance for the preceding period. Performance is measured using Earned Value Management (EVM). The Project Manager is responsible for accounting for cost deviations and presenting the Project Sponsor with options for getting the project back on budget. The Project Sponsor has the authority to make changes to the project to bring it back within budget.

COST MANAGEMENT APPROACH

This project uses OpenProject as its Project Management Information System (PMIS). Costs are managed at the Work Package level, which provides the highest level of granularity and visibility into cost performance. This approach is aligned with PMBOK best practices, as the availability of a PMIS enables cost tracking down to individual tasks and features.

Control Accounts are established per sprint budget. Earned Value calculations for each sprint measure and manage the financial performance of the project. Cost estimates are detailed within individual work packages, with overall cost management accuracy maintained at the sprint budget level.

Credit for work is assigned at the work package level. Work started on a work package grants 50% credit; the remaining 50% is credited upon full completion. Costs are rounded to the nearest peso and work hours to the nearest whole hour.

Cost variances of +/- 0.1 in the CPI or SPI change the cost status to cautionary (yellow in project status reports). Variances of +/- 0.2 escalate to alert status (red), requiring corrective action from the Project Manager. All corrective actions require a formal change request approved by the Project Sponsor before implementation.

Project Assumptions & Sprint Configuration

a. Sprint Assumptions

The following assumptions form the basis for all budget and cost calculations in OpenProject for Term 3 Sprint 1:

Parameter	Value
Sprint Duration	2 weeks
Hours per week per member	10 hours
Hours per member per sprint	20 hours (10 hrs/week × 2 weeks)
Number of team members	4 members
Total planned hours	80 hours (20 hrs × 4 members)
Hourly rate (all members)	PHP 500.00 per hour
Planned labor cost	PHP 40,000.00
Planned unit costs	PHP 19,300.00
Total Planned Value (PV)	PHP 59,300.00

All team members are assigned the same hourly rate of PHP 500.00 in OpenProject under Administration → Labor Costs → Hourly Rate.

Sprint Budget Details

a. Planned Labor Costs

Each team member is allocated 20 hours over the 2-week sprint at PHP 500.00 per hour, for a total planned labor cost of PHP 40,000.00. The table below reflects the planned labor budget per member as configured in OpenProject:

Team Member	Planned Hours	Rate/Hour	Budget
Tracie Tomon	20 hrs	PHP 500.00	PHP 10,000.00
Jenine Elaine Dulay	20 hrs	PHP 500.00	PHP 10,000.00
Charijoy Cempron	20 hrs	PHP 500.00	PHP 10,000.00
Joshua Sta Ana	20 hrs	PHP 500.00	PHP 10,000.00
TOTAL	**80 hrs**		**PHP 40,000.00**

b. Planned Unit Costs

Unit costs represent non-labor expenditures added to the sprint budget in OpenProject. The Co-working Space (Hourly) cost is set to 66 hours, matching the team's total actual Spent Time of 65.69 hours logged across all work packages. All other unit costs reflect realistic 2-week sprint consumption:

Units	Unit Name	Cost Type	Rate	Budget
66.00	hours	Co-working space (Hourly)	PHP 100.00	PHP 6,600.00
1.00	month	Internet Monthly Subscription Php1K	PHP 1,000.00	PHP 1,000.00
1.00	month	Cloud Services Monthly Subscription	PHP 100.00	PHP 100.00
1.00	package	Contingency Reserve	PHP 10,000.00	PHP 10,000.00
8.00	units	Food Allowance	PHP 100.00	PHP 800.00
8.00	units	Transportation expenses Php 100	PHP 100.00	PHP 800.00
		TOTAL PLANNED UNIT COSTS		**PHP 19,300.00**

Notes on key unit cost decisions:

- Co-working space (66 hrs): Set to match actual Spent Time logged (65.69 hrs rounded up). At PHP 100/hr, this yields PHP 6,600.00 planned vs PHP 6,568.50 actual, a near-exact match.
- Contingency Reserve (1 package × PHP 10,000): Represents approximately 25% of the labor budget, serving as a buffer for unforeseen costs within the sprint.
- Food Allowance & Transportation (8 units each): Based on 2 team meetings per week × 2 weeks × 4 members = 16 total instances, split equally between meals and transport at PHP 100 each.
- Internet and Cloud subscriptions are billed monthly, so 1 unit covers the full 2-week sprint period.

c. Actual Unit Costs — Co-working Space (Hourly)

The following table shows actual co-working space costs logged per work package in OpenProject, matched against Spent Time entries for each task and feature:

Work Package	Units	Actual Cost
Task #6327: 6.1.1 Audit and identify redundant columns...	0.69	PHP 68.50
Task #5270: Task 1 Improve LLM prompt handling	4.00	PHP 400.00
Task #5271: Task 2 Improve follow-up retrieval behavior	4.00	PHP 400.00
Task #6328: 6.1.2 Remove identified redundant columns...	3.00	PHP 300.00
Feature #4690: Feature 4.0 Display Philippine Std Time	4.00	PHP 400.00
Feature #4693: Feature 6.0 UI Decluttering (Feedback)	4.00	PHP 400.00
Feature #4696: Feature 8.0 Failed Query Keyword Tracking	16.00	PHP 1,600.00
Task #6336: Task 2 Dropdown Refinement (Frontend)	3.00	PHP 300.00
Task #6335: Task 1 UI/Table Update (Column Merger)	3.00	PHP 300.00
Task #6300: Task 1 Implement backend logic...	4.00	PHP 400.00
Task #6306: Task 1 Implement frontend pagination...	1.00	PHP 100.00
Feature #4689: Feature 3.0 Welcome Page & Form UI...	4.00	PHP 400.00
Feature #4694: Feature 7.0 Data Pagination	2.00	PHP 200.00
Feature #4674: Feature 1.0 User Account Registration	5.00	PHP 500.00
Feature #4679: Feature 2.0 LLM Source Accuracy...	8.00	PHP 800.00
TOTAL ACTUAL CO-WORKING COSTS	**65.69**	**PHP 6,568.50**

MEASURING PROJECT COSTS

a. Overview

Performance of this project is measured using Earned Value Management (EVM). EVM integrates scope, schedule, and cost baselines to provide an objective measure of project performance. OpenProject captures Spent Time and unit costs, which feed directly into the EVM calculations below.

The following EVM metrics are tracked and reported for this sprint:

- Planned Value (PV) — the authorized budget assigned to all scheduled work for the sprint
- Earned Value (EV) — the budgeted value of work performed, based on percent complete
- Actual Cost (AC) — the realized cost incurred for work performed, extracted from OpenProject Spent Time
- Cost Variance (CV = EV − AC) — positive value means under budget
- Schedule Variance (SV = EV − PV) — positive value means ahead of schedule
- Cost Performance Index (CPI = EV / AC) — greater than 1.0 means under budget
- Schedule Performance Index (SPI = EV / PV) — equal to 1.0 means on schedule

b. EVM Computation — Term 3 Sprint 1

The computations below are based on actual OpenProject data. Sprint 1 ran from March 23 to April 3, 2026, and has been fully completed, all work packages were delivered. EV equals PV because 100% of planned work was accomplished. Actual Cost (AC) of PHP 46,568.50 is composed of labor costs (80 hours × PHP 500.00 = PHP 40,000.00) plus co-working space costs logged in OpenProject (PHP 6,568.50).

EVM Metric	Interpretation	Value
Planned Value (PV)	Total authorized budget for Sprint 1 (Mar 23 – Apr 3)	**PHP 59,300.00**
Earned Value (EV)	All work packages completed and delivered — 100%	**PHP 59,300.00**
Actual Cost (AC)	Labor (80 hrs × PHP 500) + Co-working (PHP 6,568.50)	**PHP 46,568.50**

Cost Variance (CV = EV – AC)	Under budget — spent less than the value earned	**PHP 12,731.50**
Schedule Variance (SV = EV – PV)	Perfectly on schedule — all planned work delivered	**PHP 0.00**
CPI (EV / AC)	Cost efficient — PHP 1.27 earned per PHP 1.00 spent	**1.2734**
SPI (EV / PV)	Perfectly on schedule — sprint completed as planned	**1.0000**

c. EVM Interpretation

Cost Variance (CV) = PHP 12,731.50: The sprint finished under budget. The team delivered PHP PHP 59,300.00 worth of work while spending only PHP 46,568.50. The savings of PHP 12,731.50 came primarily from efficient time usage and co-working space costs coming below the planned 66-hour allocation.

Schedule Variance (SV) = PHP 0.00: The sprint was completed exactly on schedule. All planned work packages were delivered by April 3, 2026 — the end of the 2-week sprint window. A CV of zero confirms there were no delays.

CPI = 1.2734: For every PHP 1.00 spent, the team delivered PHP 1.27 of value. This strong CPI reflects disciplined cost management throughout the sprint and places the project comfortably in the green zone.

SPI = 1.0000: A perfect SPI of 1.0000 confirms the sprint was completed exactly as scheduled. All features and tasks defined in the sprint scope were delivered within the planned 2-week period (March 23 – April 3, 2026).

Cost Performance Thresholds

The following thresholds define when escalation and corrective action are required:

Metric	Yellow (Caution)	Red (Alert)
SPI	0.8 – 0.9 or 1.1 – 1.2	< 0.8 or > 1.2
CPI	0.8 – 0.9 or 1.1 – 1.2	< 0.8 or > 1.2

Yellow range (0.8–0.9 or 1.1–1.2): The Project Manager must report the reason for the variance at the next status meeting. No formal change request is required but monitoring is intensified.

Red range (< 0.8 or > 1.2): The Project Manager must report the reason and provide a detailed corrective action plan to management within five business days. The current CPI of 1.27 is above 1.2 in the positive direction, meaning the sprint finished under budget, a favorable outcome requiring no corrective action.

Reporting Format

Cost performance reporting is included in the Sprint Review Report and the Monthly Project Status Report. Each report contains a dedicated Cost Management section with the following

- All seven EVM metrics: PV, EV, AC, CV, SV, CPI, and SPI with current values
- Status color indicators (Green / Yellow / Red) per the performance thresholds above
- Commentary on any variances outside acceptable thresholds, with planned corrective actions
- A log of any Change Requests triggered by cost overruns or budget adjustments
- Updated Spent Time and unit cost data exported from OpenProject

Reports are generated at the end of each sprint and presented to the Project Sponsor and relevant stakeholders.

Cost Variance Response Process

If the CPI or SPI reaches a Control Threshold (less than 0.8 or greater than 1.2 in the negative direction), a Cost Variance Corrective Action Plan is required:

- Within 5 business days of the variance first being reported: the Project Manager presents corrective action options to the Project Sponsor
- Within 3 business days of option selection: the Project Manager presents a formal Cost Variance Corrective Action Plan
- The plan details all actions needed to bring the project back within acceptable cost and schedule performance, along with how effectiveness will be measured
- Upon acceptance, the plan becomes part of the project baseline, and all affected schedules and budgets are updated in OpenProject

Corrective actions may include increasing the project budget, reducing scope or quality, re-allocating team resources, or other measures approved by the Project Sponsor.

Cost Change Control Process

The cost change control process follows the established project change request process. All changes to the project budget, including adjustments to labor rates, unit costs, or overall budget ceilings, must be submitted via a formal Change Request and approved by the Project Sponsor before implementation.

All approved changes are reflected in updated budget records in OpenProject. No budget change takes effect until the Change Request has been formally reviewed and approved by the Project Sponsor.

Project Budget Summary

The total budget for Term 3 Sprint 1 is summarized below, reflecting the corrected and verified OpenProject configuration:

Budget Item	Amount
Planned Labor Costs (80 hrs × PHP 500.00)	PHP 40,000.00
Co-working Space (66 hrs × PHP 100.00)	PHP 6,600.00
Internet Monthly Subscription (1 × PHP 1,000.00)	PHP 1,000.00
Cloud Services Monthly Subscription (1 × PHP 100.00)	PHP 100.00
Contingency Reserve (1 package × PHP 10,000.00)	PHP 10,000.00
Food Allowance (8 units × PHP 100.00)	PHP 800.00
Transportation Expenses (8 units × PHP 100.00)	PHP 800.00
TOTAL PLANNED VALUE (PV)	**PHP 59,300.00**
Actual Labor Cost (80 hrs × PHP 500.00)	PHP 40,000.00
Actual Co-working Space Cost (logged in OpenProject)	PHP 6,568.50
TOTAL ACTUAL COST (AC)	**PHP 46,568.50**
Cost Variance (CV = EV − AC) — Under Budget	**PHP 12,731.50**
Management Reserve (10% of Labor Budget — held by Sponsor)	PHP 4,000.00

The Management Reserve of PHP 4,000.00 (10% of labor budget) is held by the Project Sponsor and was not drawn upon during Sprint 1, as the team completed all work within budget. The sprint achieved a Cost Variance of +PHP 12,731.50, finishing under budget with a CPI of 1.2734.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

_____ Date: April 1, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)